

AI Exposure and Entrepreneurship*

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Abstract

This paper investigates whether artificial intelligence is changing the rate of growth-oriented entrepreneurship. In a commuting-zone-by-sector panel of U.S. workers, we employ a difference-in-differences design comparing sectors of differing AI exposure within the same commuting zone and year. We find that a one-standard-deviation increase in occupational AI exposure lowers the incorporated self-employment rate by 0.408 percentage points (11.1% of its baseline mean) while unincorporated self-employment shows no response. Because identification uses only within-commuting-zone, within-year variation across sectors, the decline cannot be explained by any shock common to a place, and it first turns significant in 2019, four years after the onset of the deep-learning era. Exposed sectors are staffed by the managerial and professional occupations from which incorporated founders are drawn, so AI reaches would-be entrepreneurs through their own task content. We conclude that AI is attenuating the entrepreneurial diffusion channel historically associated with general-purpose technologies.

JEL Codes: J24, L26, D22, O33, R12

Keywords: Artificial intelligence, entrepreneurship, self-employment, business dynamism, local labor markets

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